## Format & Expectations

Each team will complete one capstone project using a clean and reproducible analytics pipeline. The project must begin with a business decision, not with a preferred algorithm. The final recommendation must be traceable from data, assumptions, analysis, and validation.

### Required Project Package

- Reproducible code/notebook(s) with a README and clear run instructions.
- Data dictionary and source log: source URL, download date, filters, joins, and major cleaning decisions.
- Executive brief (maximum 2 pages) written for a manager or decision-maker.
- Small interactive dashboard focused on 3–5 decision-relevant KPIs and one actionable view.
- Model/analysis card summarizing target, features, validation design, assumptions, risks, ethics/bias considerations, and limitations.

### Mandatory Analytics Rules

- Define the unit of analysis, target/outcome, prediction horizon (if any), and business decision before modeling.
- Use a validation design that matches the data-generating process. Time-dependent projects must use temporal holdout or rolling validation; random train/test splits are not sufficient when they create leakage.
- Use only information that would be available at the stated decision time. Explicitly identify and remove target leakage.
- Baseline is required. Stretch work is optional and should begin only after the baseline, validation, and business interpretation are complete.
- Do not claim causal uplift, ROI, avoided incidents, or cost savings unless the data support causal identification. Scenario-based estimates must state assumptions and include sensitivity analysis.
- Quantify business impact using an explicit decision rule, budget, threshold, capacity constraint, or scenario — not only predictive accuracy.

### Suggested 6-Week Work Plan

| Week | Primary focus | Expected checkpoint |
|---|---|---|
| 1 | Business framing, data acquisition, data dictionary, initial EDA | Problem statement + data readiness review |
| 2 | Cleaning, feature design, validation plan, baseline model/analysis | Reproducible baseline |
| 3 | Advanced model/analysis and error analysis | Model comparison + diagnostics |
| 4 | Decision rules, scenario simulation, business impact analysis | Actionable decision logic |
| 5 | Robustness/sensitivity checks, dashboard, executive interpretation | Draft dashboard + recommendations |
| 6 | Final QA, reproducibility check, executive brief, presentation/demo preparation | Final submission package |

## 1. Customer Value & Promotion Targeting for an Online Retailer

| | |
|---|---|
| **Business Question** | Which customers should the retailer prioritize for retention or promotion to increase expected future margin without over-discounting? |

### Primary Dataset

- UCI Machine Learning Repository – Online Retail II. Two years of transactional data for a UK-based non-store retailer; suitable for customer-level RFM, repeat-purchase, and value modeling. UCI Online Retail II
- Recommended scope: use the full dataset or a clearly documented time window; define an observation window and a future outcome window.

### Core Tasks

- Clean transactions, identify cancellations/returns, and construct customer-level behavioral features.
- Build RFM/cohort views and a customer value segmentation.
- Predict repeat purchase and/or future customer value over a defined holdout horizon.
- Design a promotion-targeting rule and simulate expected margin under explicit response-rate and discount assumptions.

### Baseline (Required)

- RFM segmentation plus one interpretable predictive model (e.g., logistic regression for repeat purchase and/or simple CLV/future-spend regression).
- Temporal holdout evaluation: features must be computed only from the observation window.
- Promotion scenario table using at least three discount/response assumptions.

### Stretch (Optional / Bonus)

- Gradient boosting with calibration and SHAP/feature attribution.
- Budget-constrained targeting optimization or robust sensitivity analysis across response assumptions.
- Customer-level recommendation layer (e.g., retain / nurture / no-offer) with business rationale.

### Success Metrics

- Repeat-purchase model: PR-AUC/ROC-AUC plus calibration.
- Future value/spend model: MAE/RMSE (preferably on a holdout period).
- Business metric: expected incremental margin or value under the simulated targeting policy versus a simple non-targeted policy.

### Deliverables

- Customer analytics notebook/pipeline and customer-level score table.
- Executive brief: target segments, recommended policy, expected impact, and key assumptions.
- Dashboard: segment size/value, retention risk, and promotion scenario outcomes.

### Scope Guardrails

- The dataset does not contain randomized promotion treatment/control. Do not report causal uplift or observed “revenue lift vs. control.”
- Promotion effects are scenario assumptions unless the team adds a separate experimental dataset with valid treatment information.
- Do not merge unrelated complaint datasets solely to increase feature count.